

Bin Hu

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Professional Experience

- **University of Houston**, Houston, Texas, USA
 - Assistant Professor, Department of Engineering Technology, September 2022 - present
- **Old Dominion University**, Norfolk, Virginia, USA
 - Assistant Professor, Department of Engineering Technology, July 2019 - May 2022
 - Lecturer, Department of Engineering Technology, July 2017 - June 2019

Education

- Ph.D. in Electrical Engineering, University of Notre Dame (ND), USA, 2016
- M.S. in Control Science and Engineering, Zhejiang University (ZJU), China, 2010
- B.S. in Electrical Engineering and Automation, Hefei University of Technology, China, 2007

Awarded Funds

At University of Houston: I serve as PI or UH site PI on three federal awards with a combined project value of approximately **\$1.84M** and an attributable PI/UH share of approximately **\$1.16M**, in addition to an NVIDIA Academic Grant Program in-kind equipment award.

Agency	Project	Duration	Award	Role / Share
NSF	CHS: Small: AI-Human Collaboration in Autonomous Vehicles for Safety and Security	2022–2026	\$350,961	UH site PI; 50% (\$175,481)
NASA	Adaptive Methods of Human-Autonomy Responsibility Allocation for Scalable and Safe $m:N$ Operational Architectures	2023–2026	\$750,000	UH site PI; 32% (\$240,000)
NASA STRG	Safety-Enabled and Efficient Onboard Planning for Autonomous Spacecraft via Physics-Informed Reinforcement Learning	2026–2029	\$741,899	PI; NASA Early Career Faculty (ECF25)
NVIDIA	Heterogeneous Multi-Robot Waste Detection with Conformational Runtime Monitoring	2026	In-kind	PI; 4× RTX PRO 6000 Blackwell Max-Q GPUs and 2× Jetson AGX Orin Dev Kits

At Old Dominion University: I served as PI for a \$42,000 Office of Naval Research (ONR) Rapid Solutions Learning-Projects award and as faculty mentor on two ODU Undergraduate Research Program in Cybersecurity awards totaling \$6,500. The NSF project above was originally awarded at ODU and subsequently transferred to UH when I joined UH in 2022.

Peer-Reviewed Journal Publications

1. Zhang, F., Suganda, R.R., Chen, J., Liu, W., Fu, H., **Hu, B.**, and Lin, Q., "Neural-ESO: A Dual-Pathway Architecture for Provably Robust Learning-Based Control," *IEEE Robotics and Automation Letters*, 2026.
2. Zhang, F., Chen, J., Mvogo Ahanda, J.J.B., Richter, H., Lv, G., **Hu, B.**, and Lin, Q., "Unified Disturbance-Aware Safe Kinematic Control for Closed-Architecture Robots," *Mechatronics*, vol. 119, p. 103584, 2026. doi:10.1016/j.mechatronics.2026.103584.
3. Cinun, B.C.G., Tamba, T.A., Santjoko, I.R., Wang, X., **Hu, B.**, and Gunarso, M.A., "End-to-End Design and Validation of a Low-Cost Stewart Platform With Nonlinear Estimation and Control," *Advanced Control for Applications: Engineering and Industrial Systems*, vol. 8, no. 1, e70050, 2026.
4. Tran, T., Lin, Q., and **Hu, B.**, "ELASTIC: Efficient Once For All Iterative Search for Object Detection on Microcontrollers," *IEEE Transactions on Computers*, vol. 75, pp. 2281–2289, 2026. doi:10.1109/TC.2026.3678184.
5. Garcia, K.R., Chen, J., Xiao, Y., Mishler, S., Wang, C., and **Hu, B.**, "Human Perception of AI Capabilities at Classifying Perturbed Roadway Signs," *IEEE Transactions on Human-Machine Systems*, vol. 55, no. 4, pp. 499–507, 2025. doi:10.1109/THMS.2025.3573173.
6. Chen, J.F., Zhang, F., **Hu, B.**, and Lin, Q., "Quadrotor Fault-Tolerant Control at High Speed: A Model-Based Extended State Observer for Mismatched Disturbance Rejection Approach," *IEEE Control Systems Letters*, vol. 8, pp. 2895–2900, 2024.
7. Tamba, T.A., Chandra, J., and **Hu, B.**, "Stability Analysis of a Hybrid DC-DC Buck Converter Model Using Dissipation Inequality and Convex Optimization," *Journal of Mechatronics, Electrical Power, and Vehicular Technology*, vol. 14, no. 1, pp. 47–54, 2023.
8. Garcia, K.R., Mishler, S., Xiao, Y.R., Wang, C., **Hu, B.**, Still, J.D., and Chen, J., "Drivers' Understanding of Artificial Intelligence in Automated Driving Systems: A Study of a Malicious Stop Sign," *Journal of Cognitive Engineering and Decision Making*, vol. 16, no. 4, pp. 237–251, 2022.
9. **Hu, B.** and Tamba, T.A., "Optimal Transmission Power and Controller Design for Networked Control Systems Under State-Dependent Markovian Channels," *IEEE Transactions on Automatic Control*, vol. 67, no. 10, pp. 5669–5676, 2022.
10. Li, Z.C., **Hu, B.** and Yang, Z.Y., "Co-design of Distributed Event-Triggered Controller for String Stability of Vehicle Platooning Under Periodic Jamming attacks", *IEEE Transactions on Vehicular Technology* (**IF = 6.1**), vol. 70, no.12, pp. 13115-13128, 2021.
11. J. Chen, S. Mishler and **B. Hu**, "Automation Error Type and Methods of Communicating Automation Reliability Affect Trust and Performance: An Empirical Study in the Cyber Domain", *IEEE Transactions on Human-Machine Systems* (**Regular Paper, IF=3.5**), vol. 51, no. 5, pp. 463-473, 2021.

12. **Hu, B.** "Stochastic Stability Analysis for Vehicular Networked Systems with State-dependent Bursty Fading Channels: A Self-Triggered Approach", *Automatica* (**Regular Paper, IF=4.8**), vol. 123, no. 1, 2021.
13. **Hu, B.**, "Event-based Adaptive Power Control in Vehicular Networked Systems with Bursty State-dependent Fading Channels", *IEEE Transactions on Circuits and Systems II: Express Briefs* (**IF=4**), vol. 67, no. 3, pp. 506-510, 2019.
14. **Hu, B.**, Y.B. Wang, P. Orlik, T. Koike-Akino and J.L. Guo, "Co-design of Safe and Efficient Networked Control Systems in Factory Automation with State-dependent Fading Channels", *Automatica* (**Regular Paper, IF=4.8**), vol. 105, no. 7, pp. 334-346, 2019.
15. **Hu, B.** and T. Tua, "Optimal Co-design of Industrial Networked Control Systems with State-dependent Correlated Fading Channels", *International Journal of Robust and Nonlinear Control* (**IF=3.5**), vol. 29, no.13, pp. 4472-4493, 2019.
16. Li, Z.C. and **Hu, B.**, "String Stability Analysis for Vehicle Platooning under Unreliable Communication Links with Event-Triggered Strategy", *IEEE Transactions on Vehicular Technology* (**IF = 6.1**), vol. 68, no. 3, pp. 2152-2164, 2019.
17. Chen, J., Mishler, S., **Hu, B.**, Li, N.H. and Proctor, R.W., "The Description-Experience Gap in the Effect of Warning Reliability on User Trust and Performance in a Phishing Detection Context.", *International Journal of Human-Computer Studies* (**IF=5.3**), vol. 119, no. 11, pp. 35-47, 2018.
18. **Hu, B.** and Chen, J., "Optimal Task Allocation for Human-Machine Collaborative Manufacturing Systems", *IEEE Robotics and Automation Letters (RA-L)* (**IF=4.6**), vol. 2, no. 4, pp. 1933-1940, 2017.
19. **Hu, B.** and Lemmon, M.D., "Distributed Switching Control to Achieve Almost Sure Safety for Leader-follower Vehicular Networked Systems. ", *IEEE Transactions on Automatic Control* (**Full Paper, IF=6.2**), vol. 60, no. 12, pp. 3195-3209, 2015.
20. Fei, Z.S., Liu, K.L., **Hu, B.** and Liang, J., "An Efficient Latent Variable Optimization Approach With Stochastic Constraints for Complex Industrial Process.", *Chinese Journal of Chemical Engineering*, vol. 23, no. 10, pp. 1670-1678, 2015.
21. Chi, Q. H., Zhao, Z., **Hu, B.**, Lv, Y., and Liang, J., "Multi-loop Nonlinear Internal Model Controller Design Based On A Dynamic Fuzzy Partial Least Squares Model. ", *Chemical Engineering Research and Design* (**IF=3.7**), vol. 91, no. 12, pp. 2559-2568, 2013.
22. **Hu, B.**, Z. Zhao and J. Liang. "Multi-loop Nonlinear Internal Model Controller Design Under Nonlinear Dynamic PLS Framework Using ARX-Neural Network Model." *Journal of Process Control* (**IF=3.3**), 22(1):207-217, 2012.
23. Fei, Z.S., **B. Hu**, L.B. Ye and J. Liang. "ARX-NNPLS Model Based Optimization Strategy and Its Application in Polymer Grade Transition Process." *Chinese Journal of Chemical Engineering*, 20(5): 971-979, 2012.
24. Zhao, Z., **B. Hu** and J. Liang. "Multi-loop Adaptive Internal Model Control based on a Dynamic Partial Least Squares Model." *Journal of Zhejiang University SCIENCE A*, 12(3): 190-200, 2011.

25. Duan, B., J. Liang, Z.S. Fei, M. Yang and **B. Hu**. "Nonlinear Semi-parametric Modeling Method Based on GA-ANN." *Journal of Zhejiang University-Engineering Science*, 45(6):977-983, 2011.
26. **Hu, B.**, P.Y. Zheng and J. Liang. "Multi-loop Internal Model Controller Design based on a Dynamic PLS Framework." *Chinese Journal of Chemical Engineering*, 18(2): 277–285, 2010.
27. Yang, M., **B. Hu**, Z.S. Fei, P.Y. Zheng and J. Liang. "Soft-sensing Research on the Gas Phase Ethylene Polymerization in Fluidized Bed Reactor based on DPCA-RBF Network." *Chinese Journal of Scientific Instrument*, 31(3): 481–487, 2010.
28. Fei, Z.S., **B. Hu**, L.B., Ye, P.Y., Zheng and J. Liang. "Optimal Grade Transition of Polymerization Process with Path Constraints." *Journal of Chemical Industry and Engineering (China)*, (4):893-900, 2010.
29. Liu, Y.M., L.B., Ye, P.Y., Zheng, X.R., Shi, **B. Hu** and J. Liang. "Multiscale Classification and its Application to Process Monitoring." *Journal of Zhejiang University Science C*, 11(6): 425-434, 2010.
30. Liu, Y.M., J. Liang, **B. Hu** and X.R., Shi. "A Multivariable Statistical Process Monitoring Method Based on Multiscale Analysis." *Journal of the Chemical Industry and Engineering (China)*, 60(4):952-958, 2009.

Peer-Reviewed Conference Proceedings

1. Suganda, R.R. and **Hu, B.**, "Adaptive Risk-Certified Event-Triggered Replanning for Dynamic Navigation," accepted to the *10th Conference on Robot Learning (CoRL 2026)*, Austin, TX, USA, 2026. (Acceptance Rate: 33%).
2. Suganda, R.R. and **Hu, B.**, "Formation-Aware Adaptive Conformalized Perception for Safe Leader-Follower Multi-Robot Systems," *Proceedings of the 2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026)*, Pittsburgh, PA, USA, 2026. (Acceptance Rate: 36%).
3. Tareen, S.A.K., Fan, L., Yuan, X., Lin, Q., and **Hu, B.**, "SOLAR: Switchable Output Layer for Accuracy and Robustness in Once-for-All Training," *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2026)*, Tucson, AZ, USA, 2026. (Round 2 new-submission acceptance rate: 26.88%).
4. Bautista, T., Wang, F., Govindarajan, A., Sun, L., Zhang, Y.E., Chen, J., Suganda, R.R., **Hu, B.**, Politowicz, M.S., and Chancey, E.T., "A High-Fidelity 3D Simulation Environment for Multi-Aircraft Operations," *AIAA SciTech 2026 Forum*, Orlando, FL, USA, 2026.
5. Suganda, R.R., Tran, T., Pan, M., Fan, L., Lin, Q., and **Hu, B.**, "Distributed Perception Aware Safe Leader Follower System via Control Barrier Methods," *Proceedings of the 2025 IEEE International Conference on Robotics and Automation (ICRA 2025)*, Atlanta, GA, USA, 2025. (Acceptance Rate: 38.67%).
6. Dittrich Perez, M., Sun, L., Chen, J., **Hu, B.**, Politowicz, M.S., and Chancey, E.T., "A Sigma-Point Based Uncertainty-Aware Decision Making Framework for Advanced Air Mobility," *AIAA AVIATION FORUM AND ASCEND 2025*, Las Vegas, NV, USA, 2025.

7. Zhang, Y.E., Chen, J., Sun, L., **Hu, B.**, Politowicz, M.S., and Chancey, E.T., "Task-Allocation Decisions of Human-UAS Collaboration: Effects of Workload, Trust, and Self-confidence," *Proceedings of the Human Factors and Ergonomics Society Annual Meeting*, vol. 68, no. 1, pp. 841–842, 2024.
8. Tamba, T.A. and **Hu, B.**, "An Energy Shaping-Based Control Barrier Function for Constrained Control of Robot Manipulator," *2024 14th Asian Control Conference (ASCC)*, pp. 2313–2316, Dalian, China, 2024.
9. Zhang, Y.E., Chen, J., Bautista, T., Lindsay, N., Sun, L., Ghodrati, P., **Hu, B.**, Politowicz, M.S., and Chancey, E.T., "Operator Workload and Task Allocation in $m:N$ Operational Architectures of Uncrewed Aerial Systems," *AIAA AVIATION FORUM AND ASCEND 2024*, Las Vegas, NV, USA, 2024.
10. Hernando, H., Tamba, T.A., Arthaya, B.M., and **Hu, B.**, "Design and Analysis of a Virtual Prosthetic Hand Simulator," *2023 8th International Conference on Instrumentation, Control, and Automation*, pp. 143–148, 2023.
11. Garcia, K. R., Xiao, Y., Mishler, S., Wang, C., **Hu, B.**, and Chen, J. (2021, November). Human Perception of AI Capabilities in Identifying Malicious Roadway Signs. In *TMS Proceedings 2021*. PubPub.
12. Mishler, S., Garcia, K., Fuller-Jakaitis, E., Wang, C., **Hu, B.**, Still, J., and Chen, J. "Predicting a malicious stop sign: Knowledge, exposure, trust in AI." To appear in *Proceedings of the Human Factors and Ergonomics Society 65th International Annual Meeting*. Washington DC: HFES. 2021.
13. Garcia, K., Xiao, Y., Mishler, S., Wang, C., **Hu, B.**, and Chen, J. "Human perception of AI capabilities in identifying malicious roadway signs", In *The 2021 American Psychological Association's Conference on Technology, Mind & Societ*. Nov. 3-5, 2021, Virtual.
14. **B. Hu** and T. Tamba, "Optimal Networked Control Systems with State-dependent Markov Channels", In *The 2021 American Control Conference, Louisiana, New Orleans, US, May 26-28, 2021*.
15. T. Tamba and **B. Hu**, "On A Notion of Stochastic Zeroing Barrier Function", In *The 2021 American Control Conference, Louisiana, New Orleans, US, May 26-28, 2021*.
16. T. Tamba, **B. Hu**, and Y. Nazaruddin, "On Event-Triggered Implementation of Moving Target Defense Control", *2020 International Federation of Automatic Control (IFAC-V 2020)*, Germany, July 11-17, 2020.
17. T. Tamba, **B. Hu**, and Y. Nazaruddin, "An Actuator Intrusion Detection Mechanism for Event-Triggered Moving Target Defense Control", *2019 IEEE 6th Asian Conference on Defense Technology (ACDT 2019)*, Bali, Indonesia, November 13-15, 2019.
18. **Hu, B.**, S. Hsiung and M. Kersey, "Design a Portable Sensing Platform with a Lidar and TI ARM M4 Controller", *2019 ASEE Annual Conference & Exposition (ASEE 2019)*, Tampa, Florida, June, 2019.
19. Chen, J., S. Mishler and **B. Hu**, "Conveying Automation Reliability and Automation Error Type An Empirical Study in the Cyber Domain", *Proceedings of the Human Factors and Ergonomics Society 62nd International Annual Meeting (HFES 2018)*. Philadelphia, US, October, 2018.

20. Tamba, T.A., Nazaruddin Y.Y., and **B. Hu**, "Resilient Control Under Denial-of-Service via Dynamic Event Triggering," *The 2017 Asian Control Conference (ASCC 2017)*, Gold Coast, December 17-20, 2017.
21. Samiei, A., **B. Hu**, S.Y., Zhao and L. Sun, "A Decision-Making Framework for Resilient Multi-Agents Unmanned Aircraft Systems in Risky Communication Scenarios," *AIAA Science and Technology Forum and Exposition*, Kissimmee, Florida, January, 2018.
22. **Hu, B.** and J. Chen, "Optimal Task Allocation for Human-Machine Collaborative Manufacturing Systems", *The 2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2017)*, Vancouver, Canada, September 24-28, 2017.
23. Sun, L., **B. Hu** and S.Y. Zhao. "An Event-Triggering-Based Approach for Three-Dimensional Local-Level Frame Formation Control of Leader-Follower UAVs," *The 2017 International Conference on Unmanned Aircraft Systems, ICUAS'17*, June, 2017.
24. Sabic, E., Chen, J., Mishler, S., **Hu, B.** "Recognition of Car Warnings: An Analysis of Various Alert Types," *Proceedings of 2017 ACM Conference Extended Abstracts on Human Factors in Computing Systems (CHI 17)*. **(38.7% acceptance rate)**
25. Wu, B., **B. Hu** and H. Lin. "Toward Efficient Manufacturing Systems: A Trust Based Human Robot Collaboration," *The 2017 American Control Conference*, May 24-26, Seattle, WA, USA.
26. Mishler, S., Chen, J., Sabic, E., **Hu, B.**, Li, N., and Proctor, R. W. "Description-Experience Gap: The Role of Feedback and Description in Human Trust in Automation," *The 8th International Conference on Applied Human Factors and Ergonomics (AHFE 2017)*, Los Angeles, CA, USA, July 17-21, 2017.
27. Sun, L. and **B. Hu**. "Event-triggering in Three-dimensional Leader-follower Formation Control for Unmanned Aerial Vehicles." *The ASME 2016 Dynamic Systems and Control Conference (DSCC)*, October, 2016.
28. **Hu, B.** and M.D. Lemmon. "Event triggering in Vehicular Networked Systems with Limited Bandwidth and Deep Fading." In: *Proceedings of the 53rd IEEE Annual Conference on Decision and Control (CDC 14)*, December, 2014. **(Invited Paper)**
29. **Hu, B.** and M.D. Lemmon. "Distributed Switched Supervisory Control to Achieve Almost Sure Safety for a Class of Interconnected Networked Systems." In: *Proceedings of the 11th International Conference on Control & Automation (ICCA 14)*, June, 2014.
30. **Hu, B.** and M.D. Lemmon. "Distributed Switching Control to Achieve Resilience to Deep Fades in Leader-follower Nonholonomic Systems." In: *Proceedings of the 3rd ACM International Conference on High Confidence Networked Systems (HiCoNS 14)*, April, 2014.
31. **Hu, B.** and M.D. Lemmon. "Using Channel State Feedback to Achieve Resilience to Deep Fades in Wireless Networked Control Systems." In: *Proceedings of the 2nd ACM International Conference on High Confidence Networked Systems (HiCoNS 13)*, April, 2013.
32. Li, L.C., **B. Hu** and M.D. Lemmon. "Resilient Event Triggered Systems with Limited Communication." In: *Proceedings of the 51st IEEE Annual Conference on Decision and Control (CDC 12)*, December, 2012. **(Invited Paper)**
33. **Hu, B.**, Z.S. Fei, Z. Zhao and J. Liang. "Multi-loop Nonlinear IMC Strategy Design Under PLS framework Using ARX-Neural Network Model." In: *Proceedings of 8th World Congress on Intelligent Control and Automation (WCICA 10)*, July, 2010.

34. Shi, X.R., J. Liang, L.B. Ye and **B. Hu**. "A Method of Fault Diagnosis based on PCA and Bayes Classification." In: *Proceedings of 8th World Congress on Intelligent Control and Automation (WCICA 10)*, July, 2010.

Refereed Workshop Publications and Posters

1. Santjoko, I.R., Suganda, R.R., Pan, M., and **Hu, B.**, "Distributed 3D Leader-Follower Formation Control with Field-of-View Safety via Control Barrier Functions," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026)*, Late-Breaking Results Poster, Pittsburgh, PA, USA, 2026.
2. Tran, T. and **Hu, B.**, "TrashDet: Iterative Neural Architecture Search for Efficient Waste Detection," *WASTEVISION: International Workshop on Smart Waste Monitoring at WACV 2026*, Tucson, AZ, USA, 2026.
3. Tran, T. and **Hu, B.**, "FACETS: Efficient Once-for-all Object Detection via Constrained Iterative Search," *2025 IEEE International Conference on Robotics and Automation (ICRA 2025)*, Late-Breaking Results Session, Atlanta, GA, USA, 2025.
4. Suganda, R.R. and **Hu, B.**, "Toward Embedded LLM-Guided Navigation and Object Detection for Aerial Robots," *2025 IEEE International Conference on Robotics and Automation (ICRA 2025)*, Late-Breaking Results Session, Atlanta, GA, USA, 2025.

Presentations/Seminars/Invited Talks

After joining University of Houston

1. **Invited Talk**, "Safety-enabled Learning, Control and Optimization for AI-powered Cyber-Physical Systems," *2024–2025 Assistant Professor Excellence Speaker Series (APeX)*, University of Houston, 2024–2025.
2. **Invited Talk**, "Networked Autonomous and Intelligent Learning", *TeachHouston Summer Program*, University of Houston, June, 2024.
3. **Presentation**, "Networked Autonomous and Intelligent Learning", *Central Fort Bend Chamber CTE Technology Day*, University of Houston at Sugar Land, March, 2024.
4. **Seminar**, "Networked Autonomous and Intelligent Learning", *ELET 4308 Senior Design*, University of Houston, March, 2024.
5. **Invited Talk**, "Toward Resilient Vehicular Networked Systems: Safety and Security", *Department of Electrical and Computer Engineering, University of Houston*, September, 2022.

Before joining University of Houston

1. **Invited Talk**, "Toward Resilient Vehicular Networked Systems: Safety and Security", *Department of Engineering Technology and Industrial Distribution, Texas A&M University*, March, 2022.
2. **Invited Talk**, "Toward Resilient Vehicular Networked Systems: Safety and Security", *Department of Electrical Engineering, University of North Texas*, March, 2022.

3. **Invited Talk**, "Toward Resilient Vehicular Networked Systems: Safety and Security", *Department of Engineering Technology, University of Houston*, Feb, 2022.
4. **Invited Talk**, "Toward Safe and Secure Vehicular Networked Systems", *Department of Cyber Security Engineering, George Mason University*, March, 2021.
5. **Invited Talk**, "Resilient Vehicular Networked Control System: Cybersecurity and Safety", *Department of Electrical and Computer Engineering, Old Dominion University*, January, 2021.
6. **Seminar**, "Resilient Vehicular Networked Control System: Theories and Applications", *Department of Electrical and Computer Engineering, Old Dominion University*, 2019.
7. **Invited Talk**, "Resilient Vehicular Networked Control System", *Department of Engineering Technology, Old Dominion University*, 2017.
8. **Seminar**, "Resilient Vehicular Networked Systems: Theories and Applications", *Klipsch School of Electrical & Computer Engineering, New Mexico State University*, 2016.
9. **Invited Talk**, "Event triggering in Vehicular Networked Systems with Limited Bandwidth and Deep Fading.", In: *Proceedings of the 53rd IEEE Annual Conference on Decision and Control (CDC 14)*, December, 2014.
10. **Presentation**, "Resilient Event Triggered Systems with Limited Communication", In: *the 1st Midwest Workshop on Control and Game Theory*, University of Illinois at Urbana-Champaign, April 2012.
11. **Presentation**, "Resilient Leader Follower Formation Control for Nonholonomic Systems Using Channel State Feedback", In: *the 2nd Midwest Workshop on Control and Game Theory*, University of Notre Dame, April 2013.

Supervision of Graduate and Undergraduate Students

Current Ph.D. Students

Name	Program	Joined	Primary Research Area
Richie Ryulie Suganda	ECE Ph.D.	Spring 2024	Safe learning, control, optimization, and robotic autonomy
Benedictus Christo Geroda Cinun	ECE Ph.D.	Spring 2025	Machine learning, safe reinforcement learning, and robotics
Immanuel Raynaldo Santjoko	ECE Ph.D.	Fall 2025	Trustworthy human-AI collaboration and safe autonomous systems
Vieri Akustia	ECE Ph.D.	Fall 2026	Efficient embodied AI and edge deployment

Graduate Alumni and Research Supervision at UH

- Tony Tran, M.S. Engineering Data Science, 2026. Research on efficient neural architecture search, TinyML, and embedded object detection; now Machine Learning Engineer at CFD Research Corporation.

- Pooyan Ghodrati, M.S. Engineering Data Science, 2025. NASA-supported research on AI-powered multi-drone navigation and control; now Data Scientist at Toyo Financial Group.
- Jahanvi Hitesh Dave, M.S. Computer Science, 2023. NSF-supported research on robust TinyML; now at Goldman Sachs.

Selected Undergraduate Research and Capstone Supervision at UH

- 2025: Victor Le, Angel Gomez, Hieu Le, and Nicolas Upshaw — “LLM-Based Drone Navigation and Control.”
- 2024–2025: Yosef Abushaaban, Muhammad Ansari, Damian Castillo, Asaad Elsayed, and Alejandro Zamarron — “Machine-Learning-Based Car Racing.”
- 2024: Javier Caudle, Steven Comayagua, Leandra Guzman, Weijie Huang, and Edward Perez — “Advanced Surveillance Drone.”
- 2024: Tony Tran — “Efficient Neural Architecture Search for Embedded Machine Learning,” NSF-supported undergraduate research.

At Old Dominion University:

1. Victor Ortiz (Navy), EET program (ODU), advised, graduated in December 2018. Senior Project: "Facial Detection on a Raspberry Pi 3 Robotic Neck"
2. Matthew Kersey (Coast Guard), EET program (ODU), co-advised, graduated in May 2019. Group Senior Project: "Design an Embedded Portable Platform of Lidar and Peripherals for Unmanned Aerial Systems", funded by 2019 RSLP ONR project at ODU.
3. Bryan Holland (Coast Guard), EET program (ODU), co-advised, graduated in May 2019. Group Senior Project: "Design an Embedded Portable Platform of Lidar and Peripherals for Unmanned Aerial Systems", funded by 2019 RSLP ONR project at ODU.
4. Thomas Sexton, EET program (ODU), advised, graduated in May 2019. Group Senior Project: "Design an Embedded Portable Platform of Lidar and Peripherals for Unmanned Aerial Systems", funded by 2019 RSLP ONR project at ODU.
5. Zachariah Garvin, EET program (ODU), advised, graduates in May 2019. Group Senior Project: "Design an Embedded Portable Platform of Lidar and Peripherals for Unmanned Aerial Systems", funded by 2019 RSLP ONR project at ODU.
6. Teresa Trinh (Female, Minority), MET program (ODU), co-advised, graduates in May 2019. Group Senior Project: "Design an Embedded Portable Platform of Lidar and Peripherals for Unmanned Aerial Systems", funded by 2019 RSLP ONR project at ODU.
7. Andrew Vecerkauskas, EET program (ODU), advised, graduated in May 2020. Senior Project: "Robotic Operation Systems and LiDAR Vulnerabilities", funded by 2019 undergraduate cybersecurity research program at ODU.
8. Jeremy Sklute, EET program (ODU), advised, graduated in 2020. Senior Project: "Redesign of Wireless light system used by Monarch athletic bands in performance(s)"
9. Joshua Smith, EET program (ODU), advised, graduated in 2021. Senior Project: "LiDAR Implementation with Turtlebot", funded by NSF project.

10. Skye Taylor (Female), RA from IE (Virginia Tech), advised in 2021. Project: "Adaptive task allocation preferences in different workload scenarios in driving automation systems", funded by NSF REU project.
11. Jacob Michalick, RA from ECE (ODU), advised in 2022. Project: "Applications of TinyML on MicroControllers", funded by DoD Cybersecurity Program Research.
12. Kristof Siska (Navy), EET program (ODU), advised in 2020-2022. Project: "Robust Networked Drone Systems Against Jamming Attacks", funded by ODU Undergraduate Research Program in Cybersecurity.

Teaching

At University of Houston:

- Instructor, ELET2300: Introduction to C++, Fall 2022, Spring 2023, Fall 2023, Fall 2024.
- Instructor, ELET4421: Computer Networks, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026, Fall 2026.
- Instructor, ELET6316: Network Routing Algorithms and Protocols, Spring 2025.
- Instructor, ELET2305: Semiconductor Devices and Circuits, Fall 2025.
- **Co-Instructor**, ELET4397: AI in Robotics, Spring 2026.
- Instructor, ELET4397: AI in Embedded System, Fall 2026.
- Reformed ELET2300 (Introduction to C++) with new lecture, quiz, laboratory, and assessment materials aligned with the departmental curriculum.
- Reconstructed ELET4421 (Computer Networks) to integrate lecture and hands-on networking laboratories using Cisco Packet Tracer.
- Developed/co-developed new AI-focused selected-topics courses integrating machine learning, robotics, embedded AI, and hands-on laboratory activities.

Selected University of Houston Teaching Evaluations (5-point scale; higher is better). The section average is the aggregate instructor score reported in the course evaluation.

Course	Course Title	Semester	Section Avg.	Benchmark Avg.
ELET4397	AI in Robotics	Spring 2026	4.42	4.32
ELET4421	Computer Networks	Spring 2026	4.68	4.32
ELET2305	Semiconductor Devices and Circuits	Fall 2025	4.69	4.18
ELET4421	Computer Networks	Fall 2025	4.63	4.30
ELET6316	Network Routing Algorithms and Protocols	Spring 2025	4.89	4.63
ELET4421	Computer Networks	Spring 2025	4.75	4.34
ELET2300	Introduction to C++	Fall 2024	4.33	–
ELET4421	Computer Networks	Fall 2024	4.43	–
ELET4421	Computer Networks	Spring 2024	4.48	4.23
ELET2300	Introduction to C++	Fall 2023	4.26	4.22
ELET2300	Introduction to C++	Spring 2023	4.44	4.16
ELET2300	Introduction to C++	Fall 2022	4.51	4.15

At Old Dominion University:

- Instructor, EET 110, Electrical Circuits I, Spring '18, Spring '19, Spring '20
- Instructor, EET 120, Logic Circuits and Microprocessors, Fall 18', Fall 19'
- Instructor, EET 125, Logic Circuits and Microprocessors Lab, Fall 18'
- Instructor, EET 200, Electrical Circuits II, Fall 18', Fall 19', Fall 20'
- Instructor, EET 205, Electrical Circuits Lab, Fall 17', Fall 18', Fall 20'
- Instructor, EET 225, Electronics Laboratory, Spring 19'.
- Instructor, EET 305, Advanced Technical Analysis, Fall 17', Spring 18', Spring 19'.
- Instructor, EET 330, Linear Electronics, Spring 18', Summer 18', Spring 19', Summer 19', Summer 20'
- Instructor, EET 335, Linear Electronics Lab, Summer 18', Summer 19'.
- Instructor, EET 365W, Electrical Power and Machinery Laboratory, Fall 17'.
- Instructor, ENGT 434, Introduction to Senior Project, Spring 20'
- Advisor, Undergraduate Senior Project, Fall 18', Spring 19', Fall 19', Spring 20', Fall 20'.

At New Mexico State University:

- Guest Instructor, EE 501: Research Topics in ECE

At University of Notre Dame:

- Teaching Assistant, EE 47012: Audio Technology, Fall 13'
- Lab Teaching Assistant, EE 20222: Introduction to Electrical Engineering, Fall 10'.

Dissertation and Thesis

- Ph.D. Dissertation: "Stochastic Safety and Efficiency for Vehicular Networked Systems: Theories and Applications". Department of Electrical Engineering, The University of Notre Dame.

Advisor: Dr. Michael D. Lemmon.

Thesis Committee Members: Dr. Vijay Gupta, Dr. Panos Antsaklis, and Dr. Hai Lin

Awards & Honors

- **NASA Early Career Faculty Award (ECF25)**, Space Technology Research Grants Program, 2026.
- **NVIDIA Academic Grant Program Award**, "Heterogeneous Multi-Robot Waste Detection with Conformal Runtime Monitoring," 2026.
- **Assistant Professor Excellence Speaker Series (APeX)**, University of Houston, 2024–2025.
- Department Fellowship, University of Notre Dame, 2010.

- Second Prize Scholarship, Zhejiang University, 2009.
- Excellent Undergraduate Thesis Award, Hefei University of Technology, 2007.
- First Class Excellent Academic Scholarship (top 3%), Hefei University of Technology, 2006.

Service

At University of Houston:

- **Service to Department**

Sep. 2024-present:	Graduate Program Coordinator of the Network Communications Master Program
Jan. 2023-present:	Member of the Engineering Technology Scholarship Committee
Sep. 2023 and Feb. 2025 - present:	Member of Computer Engineering Technology Faculty Search Committee

- **Service to Cullen College of Engineering**

- PhD Proposal Defense Committee (Industrial Engineering): Reza Mirjalili
- September 2024: participated and showcased research project demos at 5th annual Fort Bend Chamber Technology Expo
- June 2024: presented research work in Summer Outreach Seminar for teachHOUSTON which is a preservice teachers program that helps prepare future STEM teachers in K-12 schools.
- April 2024: participated and showcased research project at the undergraduate symposium at the University of Houston
- March 2024: participated and showcased research project at the Accepted Student Welcome event.
- March 2024: participated and presented research work at Technology Day Event hosted by the Central Fort Bend Chamber of commerce.
- March 2024: participated and presented research work to about 126 high school students from Houston ISD at the University of Houston, Sugar Land campus.

Professional Service

- **Associate Editor**, IEEE International Conference on Robotics and Automation (ICRA 2027), Area: "Human-Centered Robotics and Automation."
- **Session Chair/Co-Chair**, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2026), Pittsburgh, PA, USA.
- **Co-Chair**, IEEE International Conference on Robotics and Automation (ICRA 2025), Atlanta, GA, USA.
- Member, IEEE Human-Centered AI in Transportation (HAIT) Technical Committee
- Reviewer

- *IEEE Transactions on Automatic Control, Automatica, IEEE Transactions on Automation Science and Engineering, IEEE Transactions on Cognitive Communications and Networking, IEEE Transactions on Intelligent Transportation System, IEEE Transactions on Vehicular Technology, IEEE Transactions on Industrial Informatics, IEEE Transactions on Circuits and Systems II: Express Briefs, IEEE Transactions on Control of Network Systems, Journal of Intelligent Transportation Systems: Technology, Planning and Operations, Journal of Process Control, Journal of Intelligent and Robotic Systems, Journal of Intelligent & Robotic Systems, Robotics, IEEE American Control Conference (ACC), IEEE Conference on Decision and Control (CDC), IEEE International Conference on Control & Automation (ICCA)*

Membership in Professional Societies

Institute of Electrical and Electronics Engineers (IEEE)

IEEE Robotics and Automation Society (RAS)

American Institute of Aeronautics and Astronautics (AIAA)

Association for Computing Machinery (ACM)

IEEE Intelligent Transportation Systems Society

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